Market determinants for the SPY violation rates

April 2025

The violation rates separately for calls and puts vrc_k, vrp_k have been determined for each day k from 1.Jan.25 to 31.Aug.25, for a total of 165 days.

We regress against market variables for day k:

- SPX log-return $r_k = \log(S_k/S_{k-1})$
- VIX_k
- SKEW_k
- normalized trading volumes for calls $V_{C,k}$ and puts $V_{P,k}$, defined as $V_{i,k}^{\mathrm{norm}} = \frac{1}{sd(V_i)}(V_{i,k} - \mathbb{E}[V_{i,k}])$.

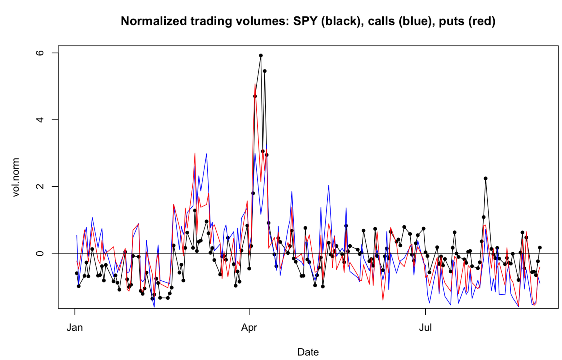

Figure 1: The normalized trading volumes.

The normalized option trading volumes are shown in Figure 1. They follow a pattern very similar to the trading volume of the underlying SPY. Also, there is no large asymmetry between the trading volumes for calls and puts.

Exploratory data analysis

The plots in Figure 2 show the violation rates (blue) and the main market variables: VIX (red), normalized option c/p trading volume (orange) and the SPY log-returns (black bars).

Some comments. There is a curious asymmetry between calls and puts violation rates.

Calls. For calls there is a large spike in violation rates in Apr-2025, roughly at the same time when the trading volume and VIX are peaking. This is related to the "tariffs day" market drop and subsequent recovery.

However, a similar large peak in calls trading volume in Mar-2025 does not lead to an increase in TP2 violation rates.

Puts. Surprisingly, there is no Apr-2025 peak in RR2 violation rates for puts. The violation rates do have a smaller peak at the end of Mar-2025 but much larger peaks in Jan and Aug, when there is no clear connection to market behavior. In Jan and Aug the VIX is at moderate levels, and the options trading volume is below average. There is no clear connection of the peaks in the RR2 violation rates to market behavior.

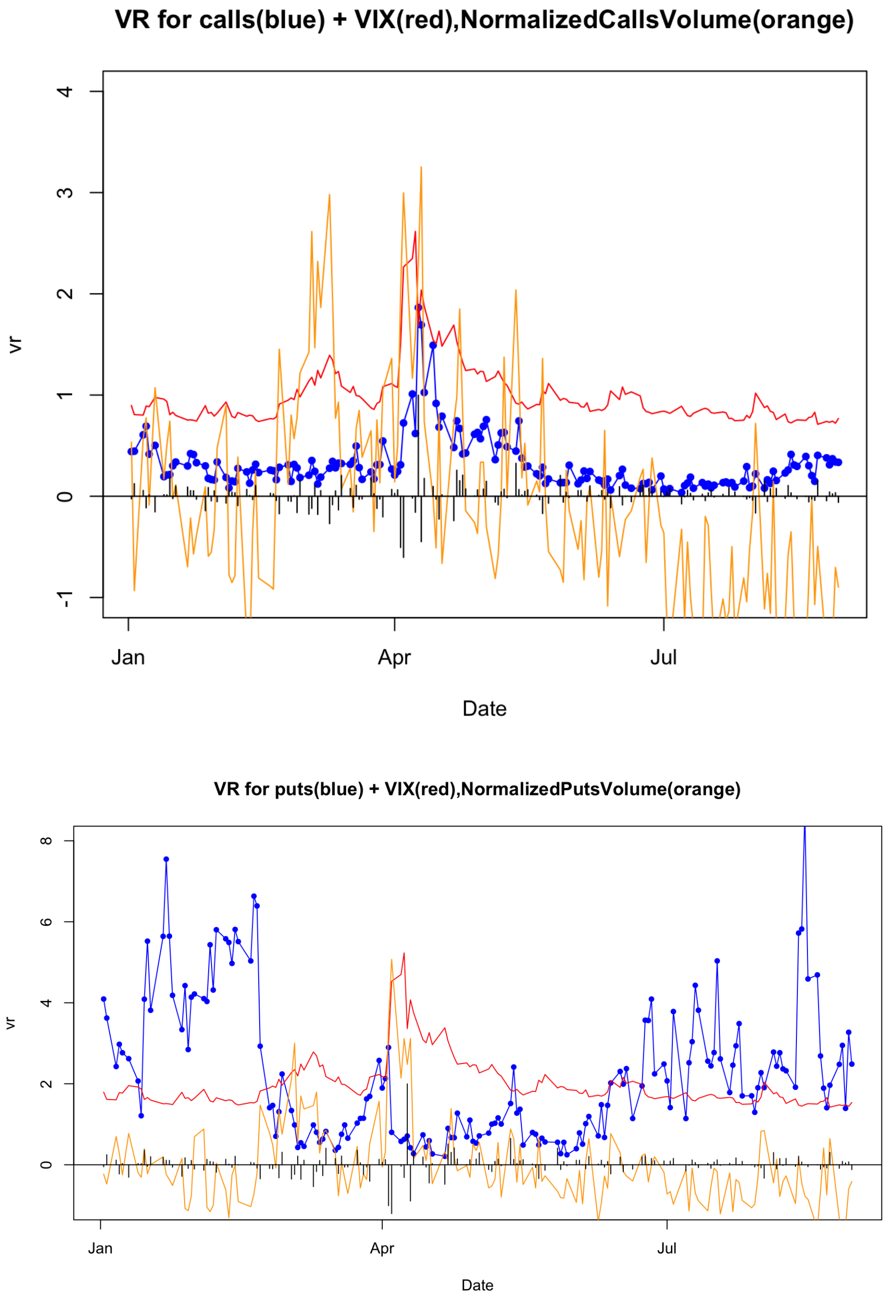

Figure 2: The TP2/RR2 violation rates for SPY options (blue) [%]. The plot shows also the normalized C/P trading volumes (orange) and the VIX (in red). The black bars show the daily SPY log-returns $r_t = \log(S_t/S_{t-1})$.

Regressions

We would like to explain the violation rates in terms of the market variables on the same day. We run two linear regressions of the form

(1) fit1: $vr_t[\%] = a + b \cdot VIX_t + \epsilon_t$

(2) fit2: $vr_t[\%] = a + b \cdot r_t + c \cdot VIX_t + d \cdot skew_t + e \cdot vol.norm_t + \epsilon_t$

We denote $r_t = \log(S_t/S_{t-1})$ the log-returns of the underlying SPY. For calls/puts we use the normalized trading volume for calls/puts separately.

The results of the regression for both calls and puts are shown in Table 1.

Some comments:

1) For calls, the most significant determinants of TP2 violation rates are SPY log-ret and VIX. The TP2 violation rates increase slightly with trading volume for calls. However, keeping only the SPY log-returns in the fit produces a very poor fit $R^2 \sim 0.02$. VIX appears to be the main determinant.

2) For puts, the main determinants are SPY log-ret (although with large errors) and VIX. The puts trading volume has a negative impact on the RR2 violation rates.

3) The calls and puts fits are dissimilar. The signs of the regression coefficients for VIX and trading volume change sign between calls and puts. Mike finds negative coefficients for VIX for both calls and put, but I get a negative coefficient only for puts.

Table 1: Coefficients for the two regressions (1) and (2) for the violation rates.

	fit1 (calls)	fit2 (calls)	fit1 (puts)	fit2 (puts)
const	(-0.254 ± 0.054)***	(-0.53 ± 0.25)*	(-2.494 ± 0.182)***	(-12.13 ± 1.54) ***
Log-ret		(5.94 ± 1.09)***		(0.919 ± 6.36)
VIX	(0.029 ± 0.003)***	(0.03 ± 0.01)***	(-0.144 ± 0.019)***	-0.023 ± 0.023
SKEW		(0.001 ± 0.001)		0.101 ± 0.008***
Vol.norm		(0.02 ± 0.02)		-0.084 ± 0.12
Adj R-sq	0.425	0.516	0.255	0.609
Nr. obs.	165	165	165	165